QUESTION -

3. **Jump Game - Minimum Jumps to Reach End**

Given an array **arr[]** of non-negative integers, where each element represents the maximum number of steps you can jump forward from that index, determine the minimum number of jumps required to reach the **last index** starting from the **first index**. If it is not possible to reach the end, return **-1.**

Examples:

Input: *arr[] = [1, 3, 5, 8, 9, 2, 6, 7, 6, 8, 9]*

Output: *3*

Explanation: *First jump from 1st element to 2nd element with value 3. From here we jump to 5th element with value 9, and from here we will jump to the last.*

Test Case	Input String	Expected Output
1	*[1, 3, 5, 8, 9, 2, 6, 7, 6, 8, 9]*	*3*
2	*[1, 4, 3, 2, 6, 7]*	*2*
3	*[0, 10, 20]*	*0*
4	*[0, 3, 10, 20]*	*0*
5	*[1, 10, 20,5,6,61]*	*2*

SOLUTION -

```python
class Solution:
    def minJumps(self, arr):
        n = len(arr)

        # If array has only one element,
        # we are already at the end.
        if n <= 1:
            return 0

        # Cannot move anywhere.
        if arr[0] == 0:
            return -1

        jumps = 0           # Number of jumps taken
        current_end = 0     # End of current jump range
        farthest = 0        # Farthest index we can reach

        # No need to process the last index
        for i in range(n - 1):

            # Update the farthest position we can reach
            farthest = max(farthest, i + arr[i])

            # Reached the end of current jump range
            if i == current_end:

                jumps += 1
                current_end = farthest

                # If we cannot move forward
                if current_end == i:
                    return -1

        return jumps
```